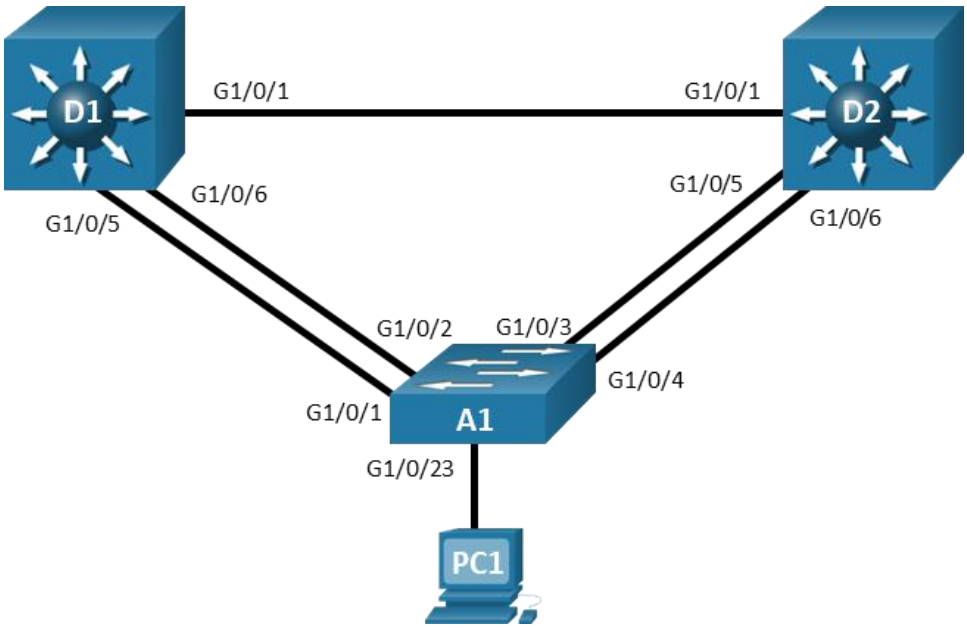
Lab - Implement RSTP and Advanced STP Modifications

Topology



Addressing Table

Device	Interface	IPv4 Address
D1	VLAN1	10.0.0.1/8
D2	VLAN1	10.0.0.2/8
A1	VLAN1	10.0.0.3/8
PC1	NIC	10.0.0.4/8

Objectives

Part 1: Build the Network and Configure Basic Device Settings

Part 2: Observe STP Convergence and Topology Change

Part 3: Configure and Verify Rapid Spanning Tree

Part 4: Implement and Observe Various Topology Tuning Methods

Part 5: Implement and Observe Various Topology Protection Mechanisms

Background / Scenario

The potential effect of a loop in the Layer 2 network is significant, as it can disrupt both connected hosts and network equipment. Spanning Tree Protocol (STP) was developed to prevent these loops, and while it works "out of the box," its default decisions do not always align with the optimal logical topology for your network. In this lab, you will first observe the operation of spanning tree protocols, including Rapid Spanning Tree

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Protocol (RSTP), to see how they protect the Layer 2 network from loops and topology disruptions. You will also work with Multiple Spanning Tree (MST), which allows you to group VLANs into separate spanning-tree instances for greater efficiency and control. You will then configure and test advanced STP features to adjust the logical topology to your requirements and explore protection mechanisms that strengthen stability and resilience. The terms "switch" and "bridge" will be used interchangeably throughout the lab.

Note: This lab is an exercise in deploying and verifying various STP mechanisms. It does not reflect networking best practices.

Note: The switches used with CCNP hands-on labs are Cisco 9300L with Cisco IOS XE release 17.12.05 (universalk9 image) and Cisco 9200L with IOS-XE release 17.12.05 (universalk9 image). Other routers and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: Ensure that the switches have been erased and have no startup configurations. If you are unsure contact your instructor.

Required Resources

- 2 Switches (Cisco 9300L with Cisco IOS XE release 17.12.05 universal image or comparable)
- 1 Switch (Cisco 9200L with Cisco IOS release 17.12.05 universal image or comparable)
- 1 PC (Windows with a terminal emulation program, such as Tera Term)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Instructions

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

In Part 1, you will set up the network topology and configure basic settings and interface addressing on routers.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram, and cable as necessary.

Step 2: Configure basic settings for each switch.

a. Console into each switch, enter global configuration mode, and apply the basic settings and interface addressing. The startup configuration is provided below for each switch in the topology.

Switch D1

```
hostname D1
spanning-tree mode pvst
banner motd # D1, Implement RSTP and Advanced STP Modifications Lab #
line con 0
exec-timeout 0 0
logging synchronous
exit
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
 exit
interface range g1/0/1, g1/0/5-6
```

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```
 switchport mode trunk
 no shutdown
 exit
vlan 2
 name SecondVLAN
 exit
interface vlan 1
 ip address 10.0.0.1 255.0.0.0
 no shut
 exit
```

Switch D2

```
hostname D2
banner motd # D2, Implement RSTP and Advanced STP Modifications Lab #
spanning-tree mode pvst
line con 0
 exec-timeout 0 0
 logging synchronous
exit
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
 exit
interface range g1/0/1, g1/0/5-6
 switchport mode trunk
 no shutdown
 exit
vlan 2
 name SecondVLAN
 exit
interface vlan 1
 ip address 10.0.0.2 255.0.0.0
 no shut
exit
```

Switch A1

```
hostname A1
banner motd # A1, Implement RSTP and Advanced STP Modifications Lab #
spanning-tree mode pvst
spanning-tree vlan 1 priority 61440
spanning-tree vlan 2 priority 61440
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
```

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```
 exit
interface range g1/0/1-4
 switchport mode trunk
 no shutdown
 exit
vlan 2
 name SecondVLAN
interface vlan 1
 ip address 10.0.0.3 255.0.0.0
 no shut
 exit
```

b. Set the clock on each switch to UTC time.

c. Save the running configuration to startup-config.

Note: Outputs and Spanning Tree topologies highlighted in this lab may be different than what you observe using your own equipment. It is critically important for you to understand how Spanning Tree makes its decisions, and how those decisions impact the operational topology of the network.

Part 2: Discover the Default Spanning Tree

Your switches have been configured and interfaces have been enabled, and the Spanning Tree Protocol, operational by default, has already converged onto a loop-free logical network. In this part of the lab, we will discover what that default spanning tree looks like and evaluate why it converged the way it did. We will do this by following the same set of steps that Spanning Tree does. We will find the Root Bridge, then find the Root Ports, and lastly see which ports are **Designated** ports, and which ports are **non-Designated** ports in our topology.

Step 1: Find the root bridge and root ports.

Our switches are running the Cisco default PVST+, and we have two VLANs in the network, so we should see two root bridges.

a. On A1, issue the command **show spanning-tree root** and observe what the output tells you about the root bridge. Amongst the lab devices being used to document this lab, A1 shows the root id with a cost of 20000 and the root port as interface GigabitEthernet 1/0/1 for both VLAN1 and VLAN2.

```
A1# show spanning-tree root

                                        Root    Hello Max Fwd
Vlan                 Root ID            Cost    Time  Age Dly  Root Port
---------------- ------------------- --------- ----- --- ---  ------------
VLAN0001         32769 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1
VLAN0002         32770 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1
```

Because we know from the physical topology diagram that A1 is connected to D1 using G1/0/1, and that interface is a GigabitEthernet interface, therefore having a cost of 20000, D1 is the root bridge for both VLAN 1 and VLAN 2. The question at this point is - why?

b. The root bridge is elected based upon which switch has the highest Bridge ID (BID). The BID is made up of a configurable priority value (which defaults to 32768) and the base MAC address for the switch. Use the command **show spanning-tree root** to gather that information from your switches to support the root bridge decision.

```
D1# show spanning-tree root
```

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```
                                        Root    Hello Max Fwd
Vlan                 Root ID            Cost    Time  Age Dly  Root Port
---------------- -------------------- --------- ----- --- ---  ------------
VLAN0001         32769 f8e9.4f9b.7f80         0    2   20  15
VLAN0002         32770 f8e9.4f9b.7f80         0    2   20  15

D2# show spanning-tree root

                                        Root    Hello Max Fwd
Vlan                 Root ID            Cost    Time  Age Dly  Root Port
---------------- -------------------- --------- ----- --- ---  ------------
VLAN0001         32769 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1
VLAN0002         32770 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1

A1# show spanning-tree root

                                        Root    Hello Max Fwd
Vlan                 Root ID            Cost    Time  Age Dly  Root Port
---------------- -------------------- --------- ----- --- ---  ------------
VLAN0001         32769 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1
VLAN0002         32770 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1
```

The first thing to look at is the priority value. It is 32768 by default. Because we are working with PVST+, a differentiator is added - the priority value is modified with the extended system ID, which is equal to the VLAN number. You can see in the output here that our three devices are using default priorities - 32769 for VLAN 1 (32768 + 1) and 32770 for VLAN 2 (32768 + 2). For each VLAN, the priority values are the same for each of the three switches. When this happens, the rest of the BID is taken into account. The rest of the BID includes the base MAC address. The **lowest** base MAC address is used to break the tie.

c. What are the base MAC addresses for the devices we are using? Issue the command **show version | include MAC** (capitalized exactly like that) on each switch.

```
D1# show version | include MAC
Base Ethernet MAC Address       : f8:e9:4f:9b:7f:80

D2# show version | include MAC
Base Ethernet MAC Address       : f8:e9:4f:9b:87:00

A1# show version | include MAC
Base Ethernet MAC Address       : 44:64:3c:e7:68:80
```

Amongst the three switches being used to document this lab, D1 has the lowest base MAC address. The OUI portion of each MAC address is the same. The Device Identifier portion of the MAC address is different. The first set of hexadecimal characters are the same. But the second set of hexadecimal characters are different; 0x7f is a lower number than 0x87. This is what caused D1 to be elected as the root bridge.

d. Each switch that is not the root will have one single root port for each VLAN. In this case, the root port is G1/0/1 for both VLANs and both A1 and D1. This port represents the lowest Path Cost to the root bridge. Path Cost is the total of the Port Costs in the path to the root bridge. Because A1 and D2 have direct connections to the root, Port cost and Path cost are the same. The Port Cost is based upon the bandwidth value of the port, and it can either be dynamically assigned or statically configured.

```
D1# show spanning-tree root
```

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```
                                        Root    Hello Max Fwd
Vlan                Root ID             Cost    Time  Age Dly  Root Port
---------------- -------------------- --------- ----- --- ---  ------------
VLAN0001         32769 f8e9.4f9b.7f80         0    2   20  15
VLAN0002         32770 f8e9.4f9b.7f80         0    2   20  15

D2# show spanning-tree root

                                        Root    Hello Max Fwd
Vlan                Root ID             Cost    Time  Age Dly  Root Port
---------------- -------------------- --------- ----- --- ---  ------------
VLAN0001         32769 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1
VLAN0002         32770 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1

A1# show spanning-tree root

                                        Root    Hello Max Fwd
Vlan                Root ID             Cost    Time  Age Dly  Root Port
---------------- -------------------- --------- ----- --- ---  ------------
VLAN0001         32769 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1
VLAN0002         32770 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1
```

Step 2: Identify Designated Ports.

The Spanning Tree Designated Port can be traced back to the early versions of the protocol, which were developed when LAN segments were shared, multiaccess networks. In these networks, there was a very real possibility that there could be users attached to a segment between two switches.

The job of the Designated Port back then was to ensure that users had a way to access the network from a given segment, and there was always one Designated Port on each segment. In the switched networks of today, there are very few shared segments, so the job of the Designated Port is more to help maintain the network topology.

A Designated Port stays active in the topology, both sending BPDUs and learning MAC addresses. Every port on the Root Bridge is a Designated Port. Further, there is one Designated Port on every segment that is not attached directly to the root.

a. On D2, issue the **show spanning-tree** command, and you will see that there are two ports now identified as being in the Designated Port role.

```
D2# show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     f8e9.4f9b.7f80
             Cost        20000
             Port        1 (GigabitEthernet1/0/1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
```

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```
  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     f8e9.4f9b.8700
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi1/0/1             Root FWD 20000     128.1    P2p
Gi1/0/5             Desg FWD 20000     128.5    P2p
Gi1/0/6             Desg FWD 20000     128.6    P2p
<output omitted>
```

b. And now look at the segments from the A1 side. Issue the **show spanning-tree** command on A1.

```
A1# show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     f8e9.4f9b.7f80
             Cost        20000
             Port        1 (GigabitEthernet1/0/1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    61441  (priority 61440 sys-id-ext 1)
             Address     4464.3ce7.6880
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi1/0/1             Root FWD 20000     128.1    P2p
Gi1/0/2             Altn BLK 20000     128.2    P2p
Gi1/0/3             Altn BLK 20000     128.3    P2p
Gi1/0/4             Altn BLK 20000     128.4    P2p
<output omitted>
```

Interfaces G1/0/3 and G1/0/4 on A1 are in the Alternate Role, which is the Cisco PVST+ version of the IEEE 802.1D Discarding role. These interfaces are up and receiving BPDUs from the Designated Ports on each segment, but they will not learn MAC addresses or forward traffic until they stop receiving those BDPUs and move to the Designated state.

c. You may have noticed in the previous output that one of the two links from A1 to D1 was not being used.

```
Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi1/0/1             Root FWD 20000     128.1    P2p
Gi1/0/2             Altn BLK 20000     128.2    P2p
```

Each switch can only have a single root port. In this example, G1/0/2, which is in the Alternate Role, would only take over if G1/0/1 were to fail. The decision about which interface to use in this scenario is based on the lowest port priority, which defaults to 128.*interface_id*.

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Part 3: Implement and Observe Rapid Spanning Tree Protocol

In Part 3, you will implement Rapid Spanning Tree Protocol (RSTP) on all the switches. Using the same basic rules, RSTP speeds up convergence significantly.

a. On D2, issue the **debug spanning-tree events** command, and then issue the **shutdown** command for interface G1/0/1 and observe the output.

```
D2# debug spanning-tree events
Spanning Tree event debugging is on
D2# conf t
Enter configuration commands, one per line.  End with CNTL/Z.
D2(config)# interface g1/0/1
D2(config-if)# shutdown
D2(config-if)#
*Sep 24 17:44:51.075: %LINK-5-CHANGED: Interface GigabitEthernet1/0/1, changed state
to administratively down
*Sep 24 17:44:52.076: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet1/0/1, changed state to down
*Sep 24 17:45:07.082: STP: VLAN0002 heard root 32770-f8e9.4f9b.7f80 on Gi1/0/5
*Sep 24 17:45:07.082:     supersedes 32770-f8e9.4f9b.8700
*Sep 24 17:45:07.082: STP: VLAN0002 new root is 32770, f8e9.4f9b.7f80 on port Gi1/0/5,
cost 40000
*Sep 24 17:45:07.083: STP: VLAN0002 sent Topology Change Notice on Gi1/0/5
*Sep 24 17:45:07.083: STP[2]: Generating TC trap for port GigabitEthernet1/0/6
*Sep 24 17:45:07.083: STP: VLAN0002 Gi1/0/6 -> blocking
*Sep 24 17:45:08.508: STP: VLAN0001 heard root 32769-f8e9.4f9b.7f80 on Gi1/0/5
*Sep 24 17:45:08.509:     supersedes 32769-f8e9.4f9b.8700
*Sep 24 17:45:08.509: STP: VLAN0001 new root is 32769, f8e9.4f9b.7f80 on port Gi1/0/5,
cost 40000
*Sep 24 17:45:08.509: STP: VLAN0001 sent Topology Change Notice on Gi1/0/5
*Sep 24 17:45:08.509: STP[1]: Generating TC trap for port GigabitEthernet1/0/6
*Sep 24 17:45:08.509: STP: VLAN0001 Gi1/0/6 -> blocking
D2(config-if)#
```

From the above output, you can see that it took a total of about 17 seconds for spanning tree to adjust to the topology change. Rapid Spanning Tree can adjust much faster. On D2, turn off debugging and reactivate the G1/0/1 interface.

```
D2(config-if)# do undebug all
All possible debugging has been turned off
D2(config-if)# no shutdown
```

b. On D1, change the spanning tree mode to rapid-pvst:

```
D1(config)# spanning-tree mode rapid-pvst
Warning: Changing STP mode can disrupt the traffic and make system unstable
Recommend to change STP mode only during maintenance window
D1(config)#
*Sep 24 17:51:19.206: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed
state to down
D1(config)#
*Sep 24 17:51:49.205: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed
state to up
D1(config)#
```

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From the above output, you can see that it took more than 30 seconds for interface VLAN1 to come back up. For an SVI (Switched Virtual Interface) to be operational, two conditions must be met:

1) The VLAN associated with the SVI must exist on the switch.

2) The VLAN must be in spanning tree forwarding mode on at least one interface.

Both conditions must be true for the SVI to function properly. It took 30 seconds to adjust. What happened to “rapid”?

c. On D1, issue the command **show spanning-tree**.

```
D1# show spanning-tree

VLAN0001
  Spanning tree enabled protocol rstp
  Root ID     Priority    32769
              Address     f8e9.4f9b.7f80
              This bridge is the root
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32769  (priority 32768 sys-id-ext 1)
              Address     f8e9.4f9b.7f80
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi1/0/1             Desg FWD 20000     128.1    P2p Peer(STP)
Gi1/0/5             Desg FWD 20000     128.5    P2p Peer(STP)
Gi1/0/6             Desg FWD 20000     128.6    P2p Peer(STP)
<output omitted>
```

The type values tell the story. Rapid spanning tree is backwards compatible with common spanning tree. It achieves this backwards compatibility by falling back to using the timers and settings of common spanning tree. In other words, we will not see the benefits of rapid spanning tree if only one switch is running it.

d. On D2 and A1, change the spanning tree mode to rapid spanning tree. On A1, first issue the **debug spanning-tree events** command, and then change the mode.

```
D2(config)# spanning-tree mode rapid-pvst


A1# debug spanning-tree events
A1# config t
A1(config)# spanning-tree mode rapid-pvst
*Sep 24 11:22:24.436: setting bridge id (which=3) prio 61441 prio cfg 61440 sysid 1
(on) id F001.4464.3ce7.6880
*Sep 24 11:22:24.437: RSTP(1): initializing port Gi1/0/1
*Sep 24 11:22:24.438: RSTP(1): Gi1/0/1 is now designated
*Sep 24 11:22:24.438: RSTP(1): initializing port Gi1/0/2
*Sep 24 11:22:24.438: RSTP(1): Gi1/0/2 is now designated
*Sep 24 11:22:24.438: RSTP(1): initializing port Gi1/0/3
*Sep 24 11:22:24.438: RSTP(1): Gi1/0/3 is now designated
*Sep 24 11:22:24.438: RSTP(1): initializing port Gi1/0/4
```

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```
*Sep 24 11:22:24.438: RSTP(1): Gi1/0/4 is now designated
*Sep 24 11:22:24.438: setting bridge id (which=3) prio 61442 prio cfg 61440 sysid 2
(on) id F002.4464.3ce7.6880
*Sep 24 11:22:24.438: RSTP(2): initializing port Gi1/0/1
*Sep 24 11:22:24.438: RSTP(2): Gi1/0/1 is now designated
*Sep 24 11:22:24.438: RSTP(2): initializing port Gi1/0/2
*Sep 24 11:22:24.438: RSTP(2): Gi1/0/2 is now designated
*Sep 24 11:22:24.438: RSTP(2): initializing port Gi1/0/3
*Sep 24 11:22:24.438: RSTP(2): Gi1/0/3 is now designated
*Sep 24 11:22:24.439: RSTP(2): initializing port Gi1/0/4
*Sep 24 11:22:24.439: RSTP(2): Gi1/0/4 is now designated
*Sep 24 11:22:24.443: RSTP(1): transmitting a proposal on Gi1/0/1
*Sep 24 11:22:24.444: RSTP(1): transmitting a proposal on Gi1/0/2
*Sep 24 11:22:24.444: RSTP(1): transmitting a proposal on Gi1/0/3
*Sep 24 11:22:24.444: RSTP(1): transmitting a proposal on Gi1/0/4
*Sep 24 11:22:24.444: RSTP(2): transmitting a proposal on Gi1/0/1
*Sep 24 11:22:24.444: RSTP(2): transmitting a proposal on Gi1/0/2
*Sep 24 11:22:24.444: RSTP(2): transmitting a proposal on Gi1/0/3
*Sep 24 11:22:24.444: RSTP(2): transmitting a proposal on Gi1/0/4
*Sep 24 11:22:25.045: RSTP(1): updt roles, received superior bpdu on Gi1/0/1
*Sep 24 11:22:25.045: RSTP(1): Gi1/0/1 is now root port
*Sep 24 11:22:25.045: RSTP(1): syncing port Gi1/0/2
*Sep 24 11:22:25.045: RSTP(1): syncing port Gi1/0/3
*Sep 24 11:22:25.045: RSTP(1): syncing port Gi1/0/4
*Sep 24 11:22:25.046: RSTP(1): updt roles, received superior bpdu on Gi1/0/2
*Sep 24 11:22:25.046: RSTP(1): Gi1/0/2 is now alternate
*Sep 24 11:22:25.046: RSTP(2): updt roles, received superior bpdu on Gi1/0/1
*Sep 24 11:22:25.046: RSTP(2): Gi1/0/1 is now root port
*Sep 24 11:22:25.046: RSTP(2): syncing port Gi1/0/2
*Sep 24 11:22:25.046: RSTP(2): syncing port Gi1/0/3
*Sep 24 11:22:25.046: RSTP(2): syncing port Gi1/0/4
*Sep 24 11:22:25.046: RSTP(2): updt roles, received superior bpdu on Gi1/0/2
*Sep 24 11:22:25.046: RSTP(2): Gi1/0/2 is now alternate
*Sep 24 11:22:25.051: RSTP(1): starting topology change timer for 35 seconds
*Sep 24 11:22:25.051: STP[1]: Generating TC trap for port GigabitEthernet1/0/1
*Sep 24 11:22:25.051: RSTP(2): starting topology change timer for 35 seconds
*Sep 24 11:22:25.051: STP[2]: Generating TC trap for port GigabitEthernet1/0/1
*Sep 24 11:22:25.052: RSTP(1): transmitting a proposal on Gi1/0/3
*Sep 24 11:22:25.052: RSTP(1): transmitting a proposal on Gi1/0/4
*Sep 24 11:22:25.052: RSTP(2): transmitting a proposal on Gi1/0/3
*Sep 24 11:22:25.053: RSTP(2): transmitting a proposal on Gi1/0/4
*Sep 24 11:22:25.060: RSTP(1): updt roles, received superior bpdu on Gi1/0/3
*Sep 24 11:22:25.060: RSTP(1): Gi1/0/3 is now alternate
*Sep 24 11:22:25.060: RSTP(1): updt roles, received superior bpdu on Gi1/0/4
*Sep 24 11:22:25.060: RSTP(1): Gi1/0/4 is now alternate
*Sep 24 11:22:25.060: RSTP(2): updt roles, received superior bpdu on Gi1/0/3
*Sep 24 11:22:25.060: RSTP(2): Gi1/0/3 is now alternate
*Sep 24 11:22:25.062: RSTP(2): updt roles, received superior bpdu on Gi1/0/4
*Sep 24 11:22:25.062: RSTP(2): Gi1/0/4 is now alternate
```

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```
*Sep 24 11:23:00.052: RSTP(1): stopping topology change timer
*Sep 24 11:23:00.052: RSTP(2): stopping topology change timerstate to up
```

A1 was the last switch that was configured for RSTP. As you can see from the debug output it took approximately .600 milliseconds for RSTP to calculate STP instances for VLAN 1 and VLAN 2. This is the “rapid” in rapid spanning tree.

e. Turn off debugging on A1.

Part 4: Implement and Observe Various Topology Tuning Methods

In Part 4, you will implement various topology tuning methods.

Step 1: Controlling the Root Bridge.

The current root bridge was elected based on the lowest Bridge ID (consisting of the Priority, extended system ID equal to the VLAN ID, and base MAC address values).

With the priority and extended system IDs being identical, the root bridge's MAC is numerically smaller than the local bridge’s MAC. The result is that in a completely un-configured network, one single switch will be elected as the root bridge. The resulting choice of switch may or may not be desirable.

There are two basic ways to manipulate the configuration to control the location of the root bridge:

- The **spanning-tree vlan** *vlan-id* **priority** *value* command can be used to manually set a priority value
- The **spanning-tree vlan** *vlan-id* **root** { **primary** | **secondary** } command can be used to automatically set a priority value.

The difference between the two is that the **priority** command will set a specific number (multiple of 4096) as the priority. This number must be an increment of 4096. The **root primary** command will set the local bridge's priority to 24,576 (if the local bridge MAC is lower than the current root bridge's MAC) or 4096 lower than the current root's priority (if the local bridge MAC is higher than the current root bridge's MAC). Notice that 24,576 is the sixth increment of 4096.

The logic behind this operation is straight-forward. The **root primary** command tries to lower the priority only as much as is needed to win the root election, while leaving priorities between 24576 and the default 32768 for use by secondary bridges. The command always takes the entire Bridge ID into account when computing the resulting priority value.

The **spanning-tree vlan** *vlan-id* **root secondary** command will statically set the local bridge’s priority to 28,672. In an otherwise unconfigured network where all switch priorities default to 32,768, the **root primary** command will set the priority on the switch to 24,576 (two 4096 increments lower than the default priority) while the **root secondary** command will set the priority on the secondary root to the 28,672 (one 4096 increment lower than the default priority).

a. Modify D1 and D2 so that D1 is elected the primary root bridge for VLAN 1 and D2 is elected the primary root bridge for VLAN 2. D1 should be elected as the secondary root bridge for VLAN 2, and D2 should be elected as the secondary root bridge for VLAN 1. You will need to make configuration changes on both D1 and D2.

```
D1(config)# spanning-tree vlan 1 root primary
D1(config)# spanning-tree vlan 2 root secondary


D2(config)# spanning-tree vlan 2 root primary
D2(config)# spanning-tree vlan 1 root secondary
```

b. After you have configured both D1 and D2, go to A1 and issue the command **show spanning-tree root**. In this output you will see the root bridges differentiated.

```
A1# show spanning-tree root
```

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```
                                        Root    Hello Max Fwd
Vlan                 Root ID            Cost    Time  Age Dly  Root Port
---------------- -------------------- --------- ----- --- ---  ------------
VLAN0001         24577 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1
VLAN0002         24578 f8e9.4f9b.8700     20000    2   20  15  Gi1/0/3
```

From the above output, you can see that the root port for VLAN 1 is G1/0/1 and the root port for VLAN 2 is G1/0/3.

Step 2: Adjust port cost values to impact root and designated port selection.

As the network is implemented right now, there are two direct paths between switch A1 and the root bridge for each VLAN. Path and port costs are evaluated to determine the shortest path to the root bridge. In the case where there are multiple equal cost paths to the root bridge, additional attributes must be evaluated. In our case, the lower interface number (for example, G1/0/1) is chosen as the Root Port, and the higher interface numbers (for example, G1/0/2) are put into a spanning tree Discarding state.

You can see which ports are blocked with the **show spanning-tree** *vlan-id* command or the **show spanning-tree blockedports** command. For now, examine VLAN 1 on D1.

a. On A1, issue the commands show spanning-tree vlan 1 and show spanning-tree blockedports.

```
A1# show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol rstp
  Root ID     Priority    24577
              Address     f8e9.4f9b.7f80
              Cost        20000
              Port        1 (GigabitEthernet1/0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    61441  (priority 61440 sys-id-ext 1)
              Address     4464.3ce7.6880
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi1/0/1             Root FWD 20000     128.1    P2p
Gi1/0/2             Altn BLK 20000     128.2    P2p
Gi1/0/3             Altn BLK 20000     128.3    P2p
Gi1/0/4             Altn BLK 20000     128.4    P2p

A1# show spanning-tree blockedports

Name                 Blocked Interfaces List
-------------------- ------------------------------------
VLAN0001             Gi1/0/2, Gi1/0/3, Gi1/0/4
VLAN0002             Gi1/0/1, Gi1/0/2, Gi1/0/4

Number of blocked ports (segments) in the system : 6
```

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As you can see, because VLAN 1 has its Root Port on G1/0/1, the Alternate Blocking Ports are G1/0/2, G1/0/3, and G1/0/4.

To manipulate which port becomes the Root Port on non-root bridges, change the port cost or port priority value. Remember that this change could have an impact on downstream switches as well.

Note: The changes you are about to implement are considered topology changes and *could* have a significant impact on the overall structure of the spanning tree in your switch network. Do not make these changes in a production network without careful planning and prior coordination.

b. On A1, shut down interfaces G1/0/1 and G1/0/2, assign a new port cost to G1/0/2, and then issue **no shutdown** to the ports.

```
A1# conf t
A1(config)# interface range g1/0/1-2
A1(config-if-range)# shutdown
A1(config-if-range)# exit
A1(config)# interface g1/0/2
A1(config-if)# spanning-tree cost 12
A1(config-if)# exit
A1(config)# interface range g1/0/1-2
A1(config-if-range)# no shutdown
A1(config-if-range)# end
```

c. Now verify that this impacts root port selection on A1 using the **show spanning-tree vlan 1** command.

```
A1# show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol rstp
  Root ID     Priority    24577
              Address     f8e9.4f9b.7f80
              Cost        12
              Port        2 (GigabitEthernet1/0/2)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    61441  (priority 61440 sys-id-ext 1)
              Address     4464.3ce7.6880
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi1/0/1             Altn BLK 20000     128.1    P2p
Gi1/0/2             Root FWD 12        128.2    P2p
Gi1/0/3             Desg BLK 20000     128.3    P2p
Gi1/0/4             Desg BLK 20000     128.4    P2p
```

From the output you can see that the root port selected by A1 for VLAN 1 is now interface G1/0/2, and the port (and root) cost is now 12.

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Step 3: Adjust port priority values to impact root port selection.

The next method to impact root port selection is configured on the root bridge itself. In our current network topology, A1 has two connections to the root bridge for VLAN 2, switch D2. The root port has been selected, in this case based on the lowest port ID. Port ID is made up of two values, labeled as Prio (Priority) and Nbr (Number).

Note: The port number is not necessarily equal to the interface ID. A switch may use any port number for STP purposes if they are unique for each port on the switch.

The port priority can be any value between 0 and 240, in increments of 16 (older switches may allow setting the priority in different increments).

a. On A1, issue the command **show spanning-tree vlan 2** and take note of the port ID values listed.

```
A1# show spanning-tree vlan 2

VLAN0002
  Spanning tree enabled protocol rstp
  Root ID     Priority    24578
              Address     f8e9.4f9b.8700
              Cost        20000
              Port        3 (GigabitEthernet1/0/3)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    61442  (priority 61440 sys-id-ext 2)
              Address     4464.3ce7.6880
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi1/0/1             Altn BLK 20000     128.1    P2p
Gi1/0/2             Altn BLK 12        128.2    P2p
Gi1/0/3             Root FWD 20000     128.3    P2p
Gi1/0/4             Altn BLK 20000     128.4    P2p
```

As expected with two equal-cost paths to the root bridge, the lower port ID was selected as the root port.

b. Modify the port priority of D2 interface G1/0/6 so that it becomes the preferred port.

```
D2(config)# interface range g1/0/5-6
D2(config-if-range)# shutdown
D2(config-if-range)# exit
D2(config)# interface g1/0/6
D2(config-if)# spanning-tree port-priority 64
D2(config-if)# exit
D2(config)# interface range g1/0/5-6
D2(config-if-range)# no shutdown
D2(config-if-range)# end
```

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c. On A1, issue the **show spanning-tree vlan 2** command and you will see that G1/0/4 is now the selected root port. This selection is based on the lower priority value of D2 interface G1/0/6. Notice that the lower priority value does not appear in any A1 output.

```
A1# show spanning-tree vlan 2

VLAN0002
  Spanning tree enabled protocol rstp
  Root ID     Priority    24578
              Address     f8e9.4f9b.8700
              Cost        20000
              Port        4 (GigabitEthernet1/0/4)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    61442  (priority 61440 sys-id-ext 2)
              Address     4464.3ce7.6880
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi1/0/1             Altn BLK 20000     128.1    P2p
Gi1/0/2             Altn BLK 12        128.2    P2p
Gi1/0/3             Altn BLK 20000     128.3    P2p
Gi1/0/4             Root FWD 20000     128.4    P2p
```

Step 4: Implement Spanning Tree PortFast.

In STP and RSTP, new ports must be checked for switching loops before entering the Forwarding state, which can take up to 30 seconds. This check is unnecessary for ports connected to end devices (like workstations, printers, or servers), because they do not create loops. In RSTP, these are called edge ports; others are non-edge ports.

Edge ports can transition to the Forwarding state immediately upon connection. Cisco's PortFast feature lets you manually designate a port as an edge port, bypassing the normal waiting period for all STP versions.

Edge ports are especially important in RSTP and MSTP. When a switch receives a Proposal on its Root port, it temporarily blocks (Discards) all non-edge Designated ports to prevent loops (the "Sync" process). Only after receiving Agreements from downstream switches do these ports return to Forwarding. If an end device is connected to a port not configured as edge (PortFast), it will be blocked for up to 30 seconds during this Sync process, causing connectivity delays.

Properly configuring ports connected to end devices as edge ports is critical for fast, reliable network convergence with RSTP. By default, Cisco switches treat all ports as non-edge ports, so manual configuration is necessary.

Let's see this in action for PC1 that is connected to A1.

a. Ensure that PC1 is turned on. You can configure it according to the **Addressing Table**, but it is not necessary for testing PortFast.

b. On A1, issue the command **debug spanning-tree events**, then issue the **no shutdown** command for interface G1/0/23, wait a few seconds, and issue the **shutdown** command. The log output will appear something like the output below.

```
A1# debug spanning-tree events
```

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```
Spanning Tree event debugging is on
A1# conf t
Enter configuration commands, one per line.  End with CNTL/Z.
A1(config)# interface g1/0/23
A1(config-if)# no shutdown
*Sep 24 13:48:05.932: RSTP(1): initializing port Gi1/0/23
*Sep 24 13:48:05.932: RSTP(1): Gi1/0/23 is now designated
*Sep 24 13:48:05.935: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:06.092: %LINK-3-UPDOWN: Interface GigabitEthernet1/0/23, changed state
to up
*Sep 24 13:48:07.091: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet1/0/23, changed state to up
*Sep 24 13:48:07.452: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:09.453: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:11.452: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:13.453: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:15.452: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:17.452: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:19.453: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:20.932: RSTP(1): Gi1/0/23 fdwhile Expired
*Sep 24 13:48:21.452: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:23.453: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:25.453: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:27.452: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:29.452: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:31.452: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:33.451: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:35.452: RSTP(1): transmitting a proposal on Gi1/0/23
*Sep 24 13:48:35.932: RSTP(1): Gi1/0/23 fdwhile Expired
*Sep 24 13:48:35.934: RSTP(1): starting topology change timer for 35 seconds
*Sep 24 13:48:35.934: STP[1]: Generating TC trap for port GigabitEthernet1/0/23
*Sep 24 13:48:36.381: RSTP(1): Gi1/0/2 received a tc ack
*Sep 24 13:49:10.932: RSTP(1): stopping topology change timer
A1(config-if)# shutdown
*Sep 24 13:49:25.676: %LINK-5-CHANGED: Interface GigabitEthernet1/0/23, changed state
to administratively down
*Sep 24 13:49:26.675: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet1/0/23, changed state to down
A1(config-if)# end
```

What you see here is the switch trying to go through the Proposal/Agreement process on G1/0/23. But there is no point in this because the device connected to G1/0/23 is an endpoint and does not understand Spanning Tree. As noted by the highlighted timestamps, this adds the potential of a 30-second delay before the host can send data, such as a DHCP request to the network.

c. On A1, configure interface G1/0/23 with the **spanning-tree portfast** command followed by the **no shutdown** command. This designates G1/0/23 as an interface that will never be connected to another switch, and therefore; it will never cause a loop in the topology, and subsequently allow that interface to go into forwarding mode immediately. Observe the output.

```
A1# conf t
Enter configuration commands, one per line.  End with CNTL/Z.
```

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```
A1(config)# interface g1/0/23
A1(config-if)# spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
 host. Connecting hubs, concentrators, switches, bridges, etc... to this
 interface  when portfast is enabled, can cause temporary bridging loops.
 Use with CAUTION

%Portfast has been configured on GigaEthernet0/23 but will only
 have effect when the interface is in a non-trunking mode.
A1(config-if)# no shutdown
A1(config-if)# exit
*Sep 24 13:55:13.555: RSTP(1): initializing port Gi1/0/23
*Sep 24 13:55:13.555: RSTP(1): Gi1/0/23 is now designated
*Sep 24 13:55:13.768: %LINK-3-UPDOWN: Interface GigabitEthernet1/0/23, changed state
to up
*Sep 24 13:55:14.767: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet1/0/23, changed state to up
```

From the output you can see that RSTP sees G1/0/23 as designated, and never sends a proposal on the interface, because of the portfast setting. The port comes up in less than 0.250 seconds.

There are two other ways to configure an interface as a portfast port; using the **switchport host** interface configuration command and using the **spanning-tree portfast default** global configuration command.

- **switchport host** not only enables portfast, but also statically sets the interface mode to access and disables aggregation protocols.
- The **spanning-tree portfast default** command sets the default state of interfaces that are configured as access ports with portfast enabled. All you must do is configure the interface with **switchport mode access** and portfast is engaged on that interface.

Verifying that a port is in portfast mode can be done by looking at the running-configuration, as shown below:

```
A1# show run | section interface GigabitEthernet1/0/23
interface GigabitEthernet1/0/23
 spanning-tree portfast
```

You can also verify portfast mode by examining spanning-tree details for the port. For example, use the **show spanning-tree interface** *interface-id* command to verify that the interface is in Edge mode, as shown below:

```
A1# show spanning-tree interface g1/0/23

Vlan                Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
VLAN0001            Desg FWD 200000    128.23   P2p Edge
```

Or, issue the command **show spanning-tree detail** | **section FastEthernet0/23**, as shown below:

```
A1# show spanning-tree detail | section GigabitEthernet1/0/23
          from GigabitEthernet1/0/23
 Port 23 (GigabitEthernet1/0/23) of VLAN0001 is designated forwarding
   Port path cost 200000, Port priority 128, Port Identifier 128.23.
   Designated root has priority 32769, address f8e9.4f9b.7f80
   Designated bridge has priority 61441, address 4464.3ce7.6880
```

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```
Designated port id is 128.23, designated path cost 12
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
The port is in the portfast mode
Link type is point-to-point by default
BPDU: sent 230, received 0
```

Part 5: Implement and Observe Various Topology Protection Mechanisms

In Part 5, you will implement and observe topology protection mechanisms such as root guard, bpdu guard, bpdu filter, and loop guard.

Step 1: Implement and observe Root Guard.

Root Guard prevents unauthorized switches from becoming the root bridge or causing a port to become a Root Port. When enabled on a port, if that port receives a superior BPDU, it discards the BPDU and places the port in the Root-Inconsistent (Blocking/Discarding) state. Unlike error-disable in port security configurations, the port is not shut down and will automatically return to normal once superior BPDUs are no longer received.

Note: BPDU Root Guard is used on ports connecting to external networks (such as customers) to ensure your own network remains the STP root bridge and to prevent the customer from taking over root switch selection or providing backup links. You should apply Root Guard only on ports facing untrusted networks. Applying Root Guard within your own trusted network is not recommended, as it can block necessary topology changes if the primary root switch or links fail, preventing proper STP convergence.

a. To illustrate the behavior of Root Guard, we will configure it on a designated port on D2 for VLAN 2. D2 is the root bridge for VLAN 2, so all trunk ports are designated, as shown in the following outputs.

```
D2# show spanning-tree root

                                        Root    Hello Max Fwd
Vlan                Root ID             Cost    Time  Age Dly  Root Port
---------------- ------------------- --------- ----- --- ---  ------------
VLAN0001         24577 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1
VLAN0002         24578 f8e9.4f9b.8700         0    2   20  15

D2# show spanning-tree detail | include VLAN0002
 VLAN0002 is executing the rstp compatible Spanning Tree protocol
 Port 1 (GigabitEthernet1/0/1) of VLAN0002 is designated forwarding
 Port 5 (GigabitEthernet1/0/5) of VLAN0002 is designated forwarding
 Port 6 (GigabitEthernet1/0/6) of VLAN0002 is designated forwarding
```

b. Go to A1 and verify what the root port for VLAN0002. It should be interface G1/0/4 because of the change in port priority we configured earlier on D2.

```
A1# show spanning-tree root | include VLAN0002
VLAN0002         24578 f8e9.4f9b.8700     20000    2   20  15  Gi1/0/4
```

c. Now on D2, add root guard to the ports connected to A1.

```
D2(config)# interface range g1/0/5-6
D2(config-if-range)# spanning-tree guard root
```

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```
*Sep 25 14:36:00.458: %SPANTREE-2-ROOTGUARD_CONFIG_CHANGE: Root guard enabled on port
GigabitEthernet1/0/5.
*Sep 25 14:36:00.459: %SPANTREE-2-ROOTGUARD_CONFIG_CHANGE: Root guard enabled on port
GigabitEthernet1/0/6.
D2(config-if-range)# exit
```

d. To verify that root guard is working, try to have A1 take over as root bridge for VLAN0002. Issue the command **spanning-tree vlan 2 priority 16384**.

```
A1(config)# spanning-tree vlan 2 priority 16384
```

e. Return to D1 and issue the command show spanning-tree vlan 2.

```
D2# show spanning-tree vlan 2

VLAN0002
  Spanning tree enabled protocol rstp
  Root ID     Priority    16386
              Address     4464.3ce7.6880
              Cost        40000
              Port        1 (GigabitEthernet1/0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    24578  (priority 24576 sys-id-ext 2)
              Address     f8e9.4f9b.8700
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi1/0/1             Root FWD 20000     128.1    P2p
Gi1/0/5             Desg BKN*20000     128.5    P2p *ROOT_Inc
Gi1/0/6             Desg BKN*20000      64.6    P2p *ROOT_Inc
```

This output has two indicators of the issue. First BKN* is short for "BROKEN", and *ROOT_Inc represents the Root Inconsistent message. A list of all STP inconsistent ports including the reason for their inconsistency can also be requested with the command **show spanning-tree inconsistentports**.

```
D2# show spanning-tree inconsistentports

Name                 Interface                     Inconsistency
-------------------- ----------------------------- ------------------
VLAN0001             GigabitEthernet1/0/5          Root Inconsistent
VLAN0001             GigabitEthernet1/0/6          Root Inconsistent
VLAN0002             GigabitEthernet1/0/5          Root Inconsistent
VLAN0002             GigabitEthernet1/0/6          Root Inconsistent

Number of inconsistent ports (segments) in the system : 4
```

f. To return things to normal, issue the command **no spanning-tree vlan 2 priority 16384** on A1 and then remove root guard on interfaces G1/0/5 and G1/0/6 of D2 with the command **no spanning-tree guard root**.

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Step 2: Implement and observe BPDU Guard.

PortFast allows a port to immediately enter the Forwarding state, which can create switching loops if two PortFast-enabled ports are accidentally or intentionally connected. Normally, if a PortFast port receives a BPDU, it reverts to a regular spanning-tree port, but under heavy load, this may not happen quickly, increasing the risk of a loop.

BPDU Guard provides extra protection by automatically shutting down (err-disabling) any port that receives an unexpected BPDU. This feature is mainly used on PortFast-enabled ports but can be enabled globally using the **spanning-tree portfast bpduguard default** command (affecting all PortFast ports) or per interface using the **spanning-tree bpduguard enable** command (regardless of PortFast status). This ensures that if a BPDU is received, the port is immediately disabled, preventing potential loops.

For this example, we will configure BPDU guard on a trunking interface that is a non-root port on A1. Configuring BPDU Guard on an interface that is intended to be a trunk is *not a recommended practice*; we are doing this just to demonstrate the functionality of the tool.

a. Verify the trunking ports and root ports on A1 using the commands **show spanning-tree root** and **show interface trunk**. From the output below, we see that interface G1/0/1 will meet the requirements for this demonstration (non-root trunk).

```
A1# show interface trunk

Port          Mode             Encapsulation  Status        Native vlan
Gi1/0/1       on               802.1q         trunking      1
Gi1/0/2       on               802.1q         trunking      1
Gi1/0/3       on               802.1q         trunking      1
Gi1/0/4       on               802.1q         trunking      1

Port          Vlans allowed on trunk
Gi1/0/1       1-4094
Gi1/0/2       1-4094
Gi1/0/3       1-4094
Gi1/0/4       1-4094

Port          Vlans allowed and active in management domain
Gi1/0/1       1-2
Gi1/0/2       1-2
Gi1/0/3       1-2
Gi1/0/4       1-2

Port          Vlans in spanning tree forwarding state and not pruned
Gi1/0/1       none
Gi1/0/2       1
Gi1/0/3       1

Port          Vlans in spanning tree forwarding state and not pruned
Gi1/0/4       1-2

A1# show spanning-tree root

                                        Root    Hello Max Fwd
Vlan                Root ID             Cost    Time  Age Dly  Root Port
```

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```
---------------- -------------------- --------- ----- --- ---  ------------
VLAN0001         24577 f8e9.4f9b.7f80        12    2   20  15  Gi1/0/2
VLAN0002         24578 f8e9.4f9b.8700     20000    2   20  15  Gi1/0/4
```

b. On A1 interface G1/0/1, issue the command **spanning-tree bpduguard enable**. As you can see, the interface is almost immediately err-disabled. Issue the **shutdown** command, remove BPDU Guard with the **no spanning-tree bpduguard enable** command, and then issue the **no shutdown** command to bring it back up. Verify the trunk is operational with the **show interface trunk** command.

```
A1(config)# interface g1/0/1
A1(config-if)# spanning-tree bpduguard enable
*Sep 25 10:24:22.526: %SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU from bridge
f8e9.4f9b.7f80 on port Gi1/0/1 with BPDU Guard enabled. Disabling port.
*Sep 25 10:24:22.555: %PM-4-ERR_DISABLE: bpduguard error detected on Gi1/0/1, putting
Gi1/0/1 in err-disable state
*Sep 25 10:24:23.555: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet1/0/1, changed state to down
*Sep 25 10:24:24.563: %LINK-3-UPDOWN: Interface GigabitEthernet1/0/1, changed state to
down
A1(config-if)# shutdown
*Sep 25 10:24:35.650: %LINK-5-CHANGED: Interface GigabitEthernet1/0/1, changed state
to administratively down
A1(config-if)# no spanning-tree bpduguard enable
A1(config-if)# no shutdown
*Sep 25 10:24:55.431: %LINK-3-UPDOWN: Interface GigabitEthernet1/0/1, changed state to
down
*Sep 25 10:24:57.954: %LINK-3-UPDOWN: Interface GigabitEthernet1/0/1, changed state to
up
*Sep 25 10:24:59.963: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet1/0/1, changed state to up
A1# show interface trunk

Port            Mode             Encapsulation  Status        Native vlan
Gi1/0/1         on               802.1q         trunking      1
Gi1/0/2         on               802.1q         trunking      1
Gi1/0/3         on               802.1q         trunking      1
Gi1/0/4         on               802.1q         trunking      1

Port            Vlans allowed on trunk
Gi1/0/1         1-4094
Gi1/0/2         1-4094
Gi1/0/3         1-4094
Gi1/0/4         1-4094

Port            Vlans allowed and active in management domain
Gi1/0/1         1-2
Gi1/0/2         1-2
Gi1/0/3         1-2
Gi1/0/4         1-2

Port            Vlans in spanning tree forwarding state and not pruned
Gi1/0/1         none
```

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```
Gi1/0/2         1
Gi1/0/3         1

Port            Vlans in spanning tree forwarding state and not pruned
Gi1/0/4         1-2
```

Step 3: Implement and observe BPDU Filter.

PortFast and BPDU Guard do not stop a switch from sending BPDUs on an interface. If you need to prevent BPDUs from being sent, use **BPDU Filter**. BPDU Filter can be set globally or per interface:

- **Global BPDU Filter** (**spanning-tree portfast bpdufilter default**): Applies only to PortFast-enabled ports. These ports send up to 11 BPDUs when they come up, then stop sending BPDUs. If the port receives a BPDU, both BPDU Filter and PortFast are disabled, and the port returns to normal spanning tree operation. Thus, global BPDU Filter mainly stops sending BPDUs but does not block receiving or processing them.
- **Interface BPDU Filter** (**spanning-tree bpdufilter enable**): When set on a specific port, the port completely stops sending and processing BPDUs. This can be used to separate a network into multiple independent STP domains, but without spanning tree protection, it is the administrator's responsibility to ensure these domains are only connected by a single link to prevent loops.

For this demonstration, we will configure BPDU filter at the interface level.

a. Find out how may BPDUs interface G1/0/23 on Switch A1 has sent using the **show spanning-tree interface g1/0/23 detail | i BPDU** command. Repeat the command several times to validate that the BPDU count is increasing.

```
A1# show spanning-tree interface g1/0/23 detail | include BPDU
   BPDU: sent 550, received 0
A1# show spanning-tree interface g1/0/23 detail | include BPDU
   BPDU: sent 552, received 0
A1# show spanning-tree interface g1/0/23 detail | include BPDU
   BPDU: sent 554, received 0
```

b. Configure the interface with BPDU filter using the **spanning-tree bpdufilter enable** command.

```
A1# config t
A1(config)# interface g1/0/23
A1(config-if)# spanning-tree bpdufilter enable
A1(config-if)# exit
A1(config)# end
```

c. Verify BPDUs are no longer being sent. Issue the command **show spanning-tree interface g1/0/23 detail | i BPDU** several times and you should see that the BPDU count is not increasing.

```
A1# show spanning-tree interface g1/0/23 detail | include BPDU
   BPDU: sent 592, received 0
A1# show spanning-tree interface g1/0/23 detail | include BPDU
   BPDU: sent 592, received 0
A1# show spanning-tree interface g1/0/23 detail | include BPDU
   BPDU: sent 592, received 0
```

d. Remove BPDU filter with the **no spanning-tree bpdufilter enable** command.

```
A1# config t
A1(config)# interface g1/0/23
A1(config-if)# no spanning-tree bpdufilter enable
```

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```
A1(config-if)# exit
A1(config)# end
```

e. Verify that BPDUs are now being sent.

```
A1# show spanning-tree interface g1/0/23 detail | include BPDU
   BPDU: sent 595, received 0
A1# show spanning-tree interface g1/0/23 detail | include BPDU
   BPDU: sent 597, received 0
A1# show spanning-tree interface g1/0/23 detail | include BPDU
   BPDU: sent 599, received 0
```

Step 4: Implement and observe Loop Guard.

Loop Guard prevents Root and Alternate ports from incorrectly becoming Designated ports if they stop receiving BPDUs. In STP, all ports, including blocked ports, listen for BPDUs to confirm the presence and status of neighboring devices. If a blocked port suddenly stops receiving BPDUs (for example, due to a unidirectional link caused by a fiber issue), it may mistakenly transition to the Forwarding state, creating a switching loop.

Loop Guard addresses this by keeping affected ports in a Loop-Inconsistent (non-forwarding) state until BPDUs are received again, preventing loops. It can be enabled globally (**spanning-tree loopguard default**) or per interface (**spanning-tree guard loop**). Loop Guard should not be used on PortFast-enabled ports.

For this example, we will configure Loop Guard on an Alternate port on A1 and then stop sending out BPDUs from the corresponding Designated port on the other end of the link.

a. Verify which ports are Alternate ports for VLAN 2 on A1 using the **show spanning-tree vlan 2** command.

```
A1# show spanning-tree vlan 2

VLAN0002
  Spanning tree enabled protocol rstp
  Root ID     Priority    24578
              Address     f8e9.4f9b.8700
              Cost        20000
              Port        4 (GigabitEthernet1/0/4)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32770  (priority 32768 sys-id-ext 2)
              Address     4464.3ce7.6880
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi1/0/1             Altn BLK 20000     128.1    P2p
Gi1/0/2             Altn BLK 12        128.2    P2p
Gi1/0/3             Altn BLK 20000     128.3    P2p
Gi1/0/4             Root FWD 20000     128.4    P2p
```

b. On A1, configure interface G1/0/1 with loop guard.

```
A1# config t
Enter configuration commands, one per line.  End with CNTL/Z.
A1(config)# interface g1/0/1
```

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```
A1(config-if)# spanning-tree guard loop
A1(config-if)# exit
A1(config)# end
```

c. On D1, configure the port connecting to G1/0/1 for bpdufilter; in this topology it is interface G1/0/5.

```
D1(config)# interface g1/0/5
D1(config-if)# spanning-tree bpdufilter enable
```

d. On A1, you should receive a SYSLOG message stating that Loop Guard has blocked port G1/0/1. Issue the command **show spanning-tree vlan 2** and you will see that G1/0/1 is broken. Issue the command **show spanning-tree inconsistentports** and you will see that G1/0/1 is loop-inconsistent.

```
A1#
*Sep 25 10:40:57.935: %SPANTREE-2-LOOPGUARD_BLOCK: Loop guard blocking port
GigabitEthernet1/0/1 on VLAN0001.
*Sep 25 10:40:58.934: %SPANTREE-2-LOOPGUARD_BLOCK: Loop guard blocking port
GigabitEthernet1/0/1 on VLAN0002.
A1# show spanning-tree vlan 2

VLAN0002
  Spanning tree enabled protocol rstp
  Root ID    Priority    24578
             Address     f8e9.4f9b.8700
             Cost        20000
             Port        4 (GigabitEthernet1/0/4)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32770  (priority 32768 sys-id-ext 2)
             Address     4464.3ce7.6880
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi1/0/1             Desg BKN*20000     128.1    P2p *LOOP_Inc
Gi1/0/2             Altn BLK 12        128.2    P2p
Gi1/0/3             Altn BLK 20000     128.3    P2p
Gi1/0/4             Root FWD 20000     128.4    P2p


A1# show spanning-tree inconsistentports

Name                 Interface                     Inconsistency
-------------------- ----------------------------- ------------------
VLAN0001             GigabitEthernet1/0/1          Loop Inconsistent
VLAN0002             GigabitEthernet1/0/1          Loop Inconsistent

Number of inconsistent ports (segments) in the system : 2
```

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e. On D1, remove the BPDU filter on interface G1/0/5.

```
D1(config)# interface g1/0/5
D1(config-if)# no spanning-tree bpdufilter enable
D1(config-if)#
```

f. On A1, you should see a SYSLOG message indicating Loop Guard has removed the block on interface G1/0/1. Remove the loop guard configuration.

```
A1#
*Sep 25 10:43:48.999: %SPANTREE-2-LOOPGUARD_UNBLOCK: Loop guard unblocking port
GigabitEthernet1/0/1 on VLAN0002.
*Sep 25 10:43:50.001: %SPANTREE-2-LOOPGUARD_UNBLOCK: Loop guard unblocking port
GigabitEthernet1/0/1 on VLAN0001.
A1# show spanning-tree inconsistentports

Name                 Interface                     Inconsistency
-------------------- ------------------------------ ------------------

Number of inconsistent ports (segments) in the system : 0

A1# conf t
Enter configuration commands, one per line.  End with CNTL/Z.
A1(config)# interface g1/0/1
A1(config-if)# no spanning-tree guard loop
A1(config-if)# end
```

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